Professional Software Engineering Processes and Human Factors 811373A: Persona cards

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Stakeholder Persona Cards

Background

The organization is migrating its systems to ML-KEM-based post-quantum cryptography. Most connections are already protected, but a number of legacy field devices do not support ML-KEM and can only connect using classical key exchange (fallback). These three stakeholders will use the dashboard the group generates. Their needs are partly in conflict.

Sarah – The Security Officer

"Which connections are still quantum-vulnerable, and how serious is that?"

- **Background:** Security engineer with a strong technical background. Understands the basics of the quantum threat very well.
- **Goal:** Know how much of the system is still exposed to future quantum attacks, and act quickly on possible downgrade attempts.
- **Decisions she makes:** Which connections or devices to investigate, block, or escalate.
- **Frustrations:** PQC readiness score numbers that hide details. Alerts without context.
- **Does not want:** A "green" status while HNDL exposure remains.
- **Position on fallback:** "Classical fallback must be disabled on all connections by the end of this quarter. Every fallback is a potential downgrade, and every fallback session is traffic that can be harvested today and decrypted later."

Michael – The PQC Migration Project Manager

"How close are we to PQC readiness, and what is blocking us?"

- **Background:** Experienced project manager with a background in industrial systems. Limited cryptography knowledge.
- **Goal:** Track migration progress and know whether the migration will be completed on time.
- **Decisions he makes:** Where to allocate the migration team's effort, and which exceptions to accept.
- **Frustrations:** Security issues that appear late and disrupt the plan.
- **Asks for:** A single readiness score he can report upwards.
- **Position on fallback:** "Fallback must stay enabled for legacy field devices until certified replacements arrive in 2028. Turning it off disconnects them from monitoring, which is a security risk in itself. Their telemetry is worthless within days, so the HNDL exposure is minimal."

Emma – The Chief Information Officer

"Do I need to worry?"

- **Background:** Business-oriented information security executive with limited technical knowledge. Reports to the board of the organizatioin every quarter.
- **Goal:** Understand quickly whether the post-quantum transition is on track and whether action is needed.
- **Decisions she makes:** Budget, priorities, and what to tell the board and key customers.
- **Frustrations:** Technical reports she cannot interpret. Surprises.
- **Asks for:** A green/amber/red status and the top three risks.
- **Position on fallback:** "Our customers must not experience outages. There is no additional budget this year, and we must meet the deadline in our key customer contract. And I want to hear about problems before the board does."